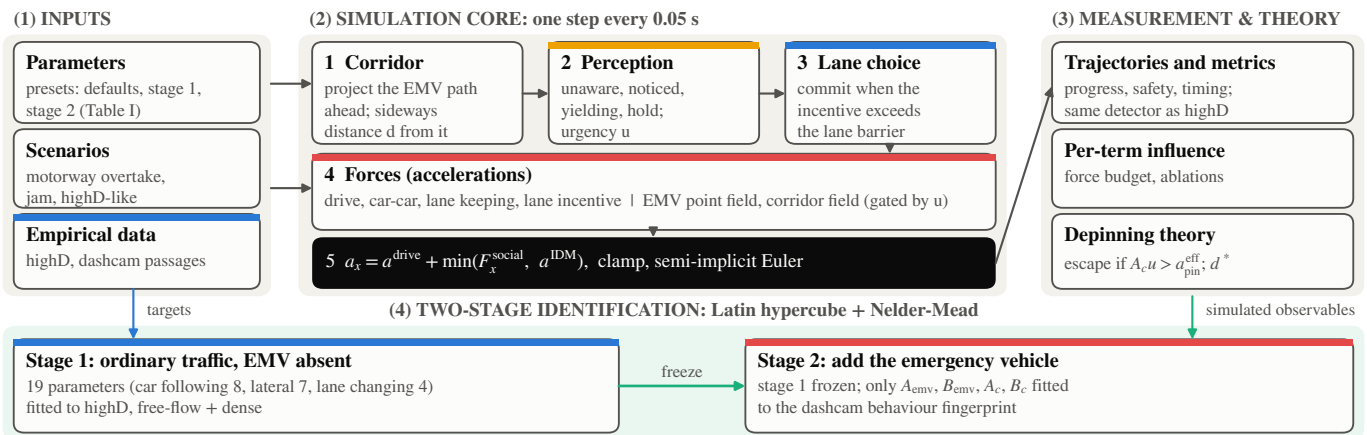
(1) INPUTS
Parameters
presets: defaults, stage 1, stage 2 (Table I)
Scenarios
motorway overtake, jam, highD-like
Empirical data
highD, dashcam passages
targets
(2) SIMULATION CORE: one step every 0.05 s
1 Corridor
project the EMV path ahead; sideways distance d from it
2 Perception
unaware, noticed, yielding, hold; urgency u
3 Lane choice
commit when the incentive exceeds the lane barrier
4 Forces (accelerations)
drive, car-car, lane keeping, lane incentive | EMV point field, corridor field (gated by u)
5 $a_x = a^{\text{drive}} + \min(F_x^{\text{social}},\ a^{\text{IDM}})$, clamp, semi-implicit Euler
(3) MEASUREMENT & THEORY
Trajectories and metrics
progress, safety, timing; same detector as highD
Per-term influence
force budget, ablations
Depinning theory
escape if $A_c u > a_{\text{pin}}^{\text{eff}}$; d^*
simulated observables
(4) TWO-STAGE IDENTIFICATION: Latin hypercube + Nelder-Mead
Stage 1: ordinary traffic, EMV absent
19 parameters (car following 8, lateral 7, lane changing 4) fitted to highD, free-flow + dense
freeze
Stage 2: add the emergency vehicle
stage 1 frozen; only A_{emv}, B_{emv}, A_c, B_c fitted to the dashcam behaviour fingerprint